CSA1718 : Artificial Intelligence

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5.Missionary And Cannibals Problem:

```
from collections import deque

def solve():
    start = (3, 3, 0)
    goal = (0, 0, 1)
    moves = [(1,0), (2,0), (0,1), (0,2), (1,1)]

    queue = deque([(start, [])])
    visited = {start}

    while queue:
        (m, c, boat), path = queue.popleft()

        if (m, c, boat) == goal:
```

```
    return path + [(m, c, boat)]

for x, y in moves:
    if boat == 0:
        nm, nc, nb = m-x, c-y, 1
    else:
        nm, nc, nb = m+x, c+y, 0

    if not (0 <= nm <= 3 and 0 <= nc <= 3):
        continue

    if ((nm > 0 and nc > nm) or
        (3-nm > 0 and 3-nc > 3-nm)):
        continue

    state = (nm, nc, nb)

    if state not in visited:
        visited.add(state)
        queue.append((state, path + [(m, c, boat)]))
```

```
for state in solve():
    print(state)
```

Output:

```
======= RESTA
(3, 3, 0)
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)
```